

MATRIX OPERATIONS TOOL USING PYTHON AND NUMPY

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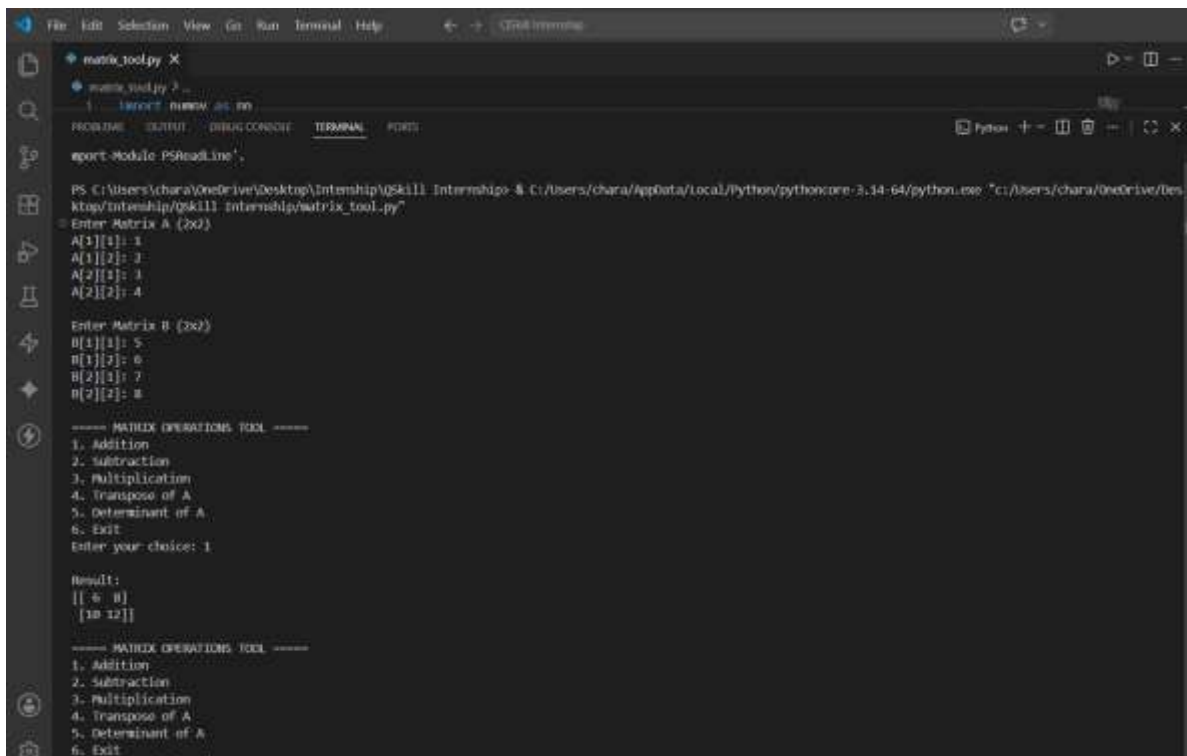
Objective :

To develop a Python application that performs matrix operations such as addition, subtraction, multiplication, transpose, and determinant calculation using NumPy.

Technologies Used :

- Python
- NumPy
- VS Code

I. Matrix Addition :



```
File Edit Selection View Go Run Terminal Help
matrix_tool.py X
1: import numpy as np
import sys
import os
import Module PSHeaderLine'

PS C:\Users\chara\OneDrive\Desktop\Internship\QSkill Internship> .& C:\Users\chara\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:\users\chara\OneDrive\Desktop\Internship\QSkill Internship\matrix_tool.py"
Enter Matrix A (2x2)
A[1][1]: 1
A[1][2]: 2
A[2][1]: 3
A[2][2]: 4

Enter Matrix B (2x2)
B[1][1]: 5
B[1][2]: 6
B[2][1]: 7
B[2][2]: 8

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 1

Result:
[[ 6  8]
 [10 12]]

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
```

- The program adds corresponding elements of two matrices and displays the resulting matrix.

II. Matrix Subtraction :

```
File Edit Selection View Go Run Terminal Help
matrix_tool.py
matrix_tool.py >
import Module 'PHeadline'.

PS C:\Users\chara\OneDrive\Desktop\Internship\Skill Internship> & C:/Users/chara/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/chara/OneDrive/Desktop/Internship/Skill Internship/matrix_tool.py"
Enter Matrix A (2x2)
A[1][1]: 1
A[1][2]: 2
A[2][1]: 3
A[2][2]: 4

Enter Matrix B (2x2)
B[1][1]: 5
B[1][2]: 6
B[2][1]: 7
B[2][2]: 8

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 2

Result:
[[-4 -4]
 [-4 -4]]

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 
```

- The program subtracts corresponding elements of Matrix B from Matrix A.

III. Matrix Multiplication :

```
File Edit Selection View Go Run Terminal Help
matrix_tool.py
matrix_tool.py >
import Module 'PHeadline'.

PS C:\Users\chara\OneDrive\Desktop\Internship\Skill Internship> & C:/Users/chara/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/chara/OneDrive/Desktop/Internship/Skill Internship/matrix_tool.py"
Enter Matrix A (2x2)
A[1][1]: 1
A[1][2]: 2
A[2][1]: 3
A[2][2]: 4

Enter Matrix B (2x2)
B[1][1]: 5
B[1][2]: 6
B[2][1]: 7
B[2][2]: 8

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 3

Result:
[[19 22]
 [41 50]]

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 
```

- The program performs matrix multiplication using NumPy's dot() function.

IV. Transpose :

```
PS c:\Users\chara\Desktop\Internship\Qskill Internship> & c:/Users/chara/AppData/Local/Python/pythoncore-3.10-64/python.exe "c:/Users/chara/Desktop/Internship/Qskill Internship/matrix_tool.py"
Enter Matrix A (2x2)
A[1][1]: 1
A[1][2]: 2
A[2][1]: 3
A[2][2]: 4

Enter Matrix B (2x2)
B[1][1]: 5
B[1][2]: 6
B[2][1]: 7
B[2][2]: 8

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 4

Result:
[[1 3]
 [2 4]]

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 
```

- The transpose operation converts rows into columns.

V. Determinant :

```
PS c:\Users\chara\Desktop\Internship\Qskill Internship> & c:/Users/chara/AppData/Local/Python/pythoncore-3.10-64/python.exe "c:/Users/chara/Desktop/Internship/Qskill Internship/matrix_tool.py"
Enter Matrix A (2x2)
A[1][1]: 1
A[1][2]: 2
A[2][1]: 3
A[2][2]: 4

Enter Matrix B (2x2)
B[1][1]: 5
B[1][2]: 6
B[2][1]: 7
B[2][2]: 8

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 5

Result:
-2.0000000000000004

===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose of A
5. Determinant of A
6. Exit
Enter your choice: 
```

- The determinant is calculated using NumPy's linalg.det() function.

Conclusion :

The Matrix Operations Tool was successfully developed using Python and NumPy. The application performs addition, subtraction, multiplication, transpose, and determinant operations through a menu-driven interface. This project helped in understanding matrix manipulation and numerical computing using Python.